

CURRICULUM VITAE

PERSONAL INFORMATION

NAME: Wookyoun Kim
DATE OF BIRTH: September 27, 1993
NATIONALITY: Korea
AFFILIATION: Korea Institute of Machinery and Materials (KIMM),
Heat Pump Research Center
POSITION: Senior Researcher
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EDUCATION

Degree	University	Department	Date
Ph.D.	KAIST	Department of Mechanical Engineering	February, 2021
Dissertation: "Study on a Thermal Network-based Model for Predicting the Thermal Resistance of Pulsating Heat Pipes"			
Advisor: Sung Jin Kim			
M.S.	KAIST	Department of Mechanical Engineering	February, 2017
Thesis: "Effect of artificial cavities on the thermal performance of pulsating heat pipes"			
Advisor: Sung Jin Kim			
B.S.	KAIST	Department of Mechanical Engineering	February, 2015

RESEARCH INTERESTS

- (1) Data center cooling (immersion cooling, direct liquid cooling, jet-impingement cooling)
- (2) High-heat-flux electronics cooling (CPU/GPU cold plate design)
- (3) Heat exchangers (PCHE for cryogenic liquid hydrogen, compact heat exchanger design)
- (4) Heat pump systems (chemisorption/adsorption heat pumps, high-temperature heat pumps)
- (5) Low GWP refrigerant systems (next-gen eco-friendly refrigerants, VLE measurement)
- (6) Two-phase heat transfer (boiling, evaporation, pulsating heat pipes, vapor chambers)
- (7) Cryogenic heat transfer (liquid hydrogen vaporizer, supercritical fluid heat transfer)

PUBLICATIONS

SCI/SCIE Journal Papers

- (1) J.S. Kim, **W. Kim**, H.S. Kim, Y. Kim, "Falling film evaporation of ammonia on smooth and Low-Fin tube arrays", *Thermal Science and Engineering Progress* 69, 104483 (2026).
- (2) B. Kim, K.H. Lee, M. Kim, S. Moon, J.W. Yoo, **W. Kim**, "Experimental and theoretical investigation of freezing phenomenon in Printed Circuit Heat Exchanger", *Applied Thermal Engineering* 273, 126455 (2025).
- (3) D.H. Kim, **W. Kim**, J.H. Kim, H.S. Kim, K.H. Lee, "Experimental studies on the VLE of R-32/R-125 and verification of experimental setup", *Journal of Mechanical Science and Technology* 38(10), 5779-5785 (2024).
- (4) H.S. Kim, D.H. Kim, J.S. Kim, **W. Kim**, Y. Kim, "A numerical study on the performance of

chemisorption heat pump according to various design conditions", *Applied Thermal Engineering* 243, 122519 (2024).

- (5) J.S. Kim, **W. Kim**, H.S. Kim, Y. Kim, "Pool boiling heat transfer of ammonia outside enhanced tubes with various fin structures", *Applied Thermal Engineering* 247, 122986 (2024).
- (6) H.S. Kim, J.H. Kim, J.S. Kim, **W. Kim**, Y. Kim, "Experimental study of a chemisorption heat pump under different operation conditions", *Applied Thermal Engineering* 240, 122274 (2024).
- (7) J. Kim, C.G. Kim, H.S. Kim, **W. Kim**, D.H. Kim, "Characterization of liquid behavior in distributor of falling-film evaporator", *Physics of Fluids* 35(6), 067124 (2023).
- (8) J.S. Kim, **W. Kim**, H.S. Kim, Y. Kim, S.H. Yoon, "Pool boiling heat transfer of ammonia outside a single tube with fin structures: Hysteresis phenomena and boiling enhancement", *International Communications in Heat and Mass Transfer* 149, 107157 (2023).
- (9) **W. Kim** and S.J. Kim, "Fundamental issues and technical problems about pulsating heat pipes", *Journal of Heat Transfer - Transactions of the ASME* 143, 100803 (2021).
- (10) **W. Kim** and S.J. Kim, "Effect of a flow behavior on the thermal performance of closed-loop and closed-end pulsating heat pipes", *International Journal of Heat and Mass Transfer* 149, 119251 (2020).
- (11) **W. Kim** and S.J. Kim, "Effect of reentrant cavities on the thermal performance of a pulsating heat pipe", *Applied Thermal Engineering* 133, 61-69 (2018).

KCI Journal Papers

- (1) J.S. Kim, D.H. Shin, J. Shim, K.H. Lee, S. Sohn, **W. Kim**, C. Song, "A Study on the Design of an Air-to-Air Plate Heat Exchanger for 300°C High Temperature Heat Pumps", *Korean Journal of Air-Conditioning and Refrigeration Engineering* 38(1), 13-25 (2026).
- (2) **W. Kim**, B. Kim, S. Sohn, K.H. Lee, J. Kim, "Experimental Investigation on the Freezing Condition of PCHE for Cryogenic Liquid Hydrogen Vaporizer", *Journal of Hydrogen and New Energy* 35(2), 240-248 (2024).
- (3) **W. Kim**, H.S. Kim, J. Kim, D.H. Kim, "An Experimental Study on The Thermal Performance Measurement and Flow Visualization of Falling Film Evaporator Using R-1233ZD(E) Refrigerant", *Korean Journal of Air-Conditioning and Refrigeration Engineering* 36(1), 9-17 (2024).
- (4) S. Sohn, **W. Kim**, "A Study on Anti-Icing Design by Conjugate Heat Transfer Analysis in a Lab-Scale PCHE for Supply of Cryogenic High Pressure Liquid Hydrogen", *Transactions of the Korean Hydrogen and New Energy Society* 33(5), 541-549 (2022).
- (5) D.H. Kim, H.S. Kim, J. Kim, **W. Kim**, "An Experimental Study on the Performance Characteristics of a Falling Film Evaporator for R-1234ze(E) Refrigerant", *Korean Journal of Air-Conditioning and Refrigeration Engineering* 35(7), 363-370 (2023).
- (6) J.S. Kim, S.A. Kim, D.H. Shin, **W. Kim**, S. Moon, Y. Chung, S. Sohn, "Comparative Study of Air Cooling and Immersion Cooling for the Thermal Management of a Cylindrical Battery Pack", *Transactions of the KSME B* 47(10), 543-550 (2023).
- (7) H.S. Kim, **W. Kim**, K.H. Lee, D.H. Kim, "Numerical Study on the Performance of Hybrid Falling Film Evaporator", *Korean Journal of Air-Conditioning and Refrigeration Engineering* 34(8), 371-379 (2022).
- (8) K.H. Lee, D.H. Kim, **W. Kim**, J.S. Kim, Y. Kim, J.W. Yoo, "Selection of the Equation of State for Thermodynamic Properties in the Low GWP Mixture Refrigerants", *Transactions of the KSME B* 46(10), 563-571 (2022).

International Conference Proceedings

- (1) 2024, "Experimental and theoretical investigation on avoiding freezing phenomena of PCHE for cryogenic liquid hydrogen vaporizer", 29th ICEC
- (2) 2024, "Understanding the thermal characteristics of falling film evaporation of R-1233ZD(E) refrigerant using flow visualization and heat transfer measurement", 11th ACRA
- (3) 2023, "Experimental investigation on the flow and thermal characteristics of falling film evaporator using R-1233ZD(E) refrigerant", 11th ICBCHT

- (4) 2019, "Experimental Investigation on the Thermal Performance of PHP Operating in a Circulation/oscillation Mode", Korea-China-Japan Joint Symposium
- (5) 2019, "Comparison of thermal performance between CEPHP and CLPHP", 4th TFEC
- (6) 2018, "Experimental investigation on the thermal performance of double-condenser pulsating heat pipes", 10th ICBCHT
- (7) 2017, "Experimental Investigation on the Thermal Performance of a Pulsating Heat Pipe with Artificial Cavities", 1st ACTS
- (8) 2017, "Experimental investigation of artificial cavities on the thermal performance of a pulsating heat pipe", InterPACK 2017

Domestic Conference Presentations

- (1) 2018, "Development of deployable radiator for spacecraft using a pulsating heat pipe", KSME 2018 Spring Annual Meeting (Thermal Engineering Division)
- (2) 2018, "Effects of circulating flow on the thermal performance of a pulsating heat pipe", KSME 2018 Annual Meeting
- (3) 2016, "Effects of artificial cavities on the startup performance of a pulsating heat pipe", KSME 2016 Annual Meeting

PATENTS

Domestic Patents (18)

- (1) Heat exchanger with anti-freezing capability (1) [2023.03.27]
- (2) Heat exchanger with anti-freezing capability (2) [2023.03.27]
- (3) Modular fin-tube reactor [2024.11.04]
- (4) Fin-tube type reactor [2024.11.04]
- (5) Tray structure for heat exchanger [2022.10.13]
- (6) Oil recovery system for refrigeration systems (1) [2022.10.13]
- (7) Oil recovery system for refrigeration systems (2) [2022.10.13]
- (8) Micro-channel reactor [2023.04.19]
- (9) Immersion cooling device [2024.10.25]
- (10) Geothermal heat supply device and heating system [2023.10.17]
- (11) Heat exchanger thermal performance testing apparatus [2023.06.16]
- (12) Ternary refrigerant composition and heat pump system [2022.10.24]
- (13) Display device [2023.10.20]
- (14) Duct assembly and combustor [2022.03.30]
- (15) Immersion cooling HVAC system and method [2022.12.22]
- (16) Heat pipe integrated reactor for adsorption heat pump [2022.04.06]
- (17) Adsorption heat pump evaporator and system [2023.03.31]
- (18) Plate pulsating heat spreader with artificial cavities [2018.02.20]

International Patents (1)

- (1) Plate pulsating heat spreader with artificial cavities (US 10,436,520 B2) [2019.10.08]

SOFTWARE REGISTRATIONS

- (1) PCHE thickness design program (C-2024-052179) [2024.12.11]
- (2) High-temperature heat pump cycle design program (C-2024-038691) [2024.10.22]
- (3) Heat exchanger HTC uncertainty analysis program (C-2023-059071) [2023.12.12]

- (4) Lab-scale heat exchanger analysis program **(C-2023-056463)** [2023.11.30]
- (5) 100 kg/hr hydrogen PCHE design program **(C-2025-056891)** [2025.12.11]
- (6) Adsorption cooling simulation program **(C-2025-034652)** [2025.09.03]
- (7) Vapor chamber performance analysis program **(C-2024-041200)** [2024.11.04]
- (8) Fin-tube heat exchanger design program **(C-2024-038692)** [2024.10.22]

TECHNOLOGY TRANSFERS

- (1) Heat pipe and vapor chamber structural design for electronics cooling
- (2) Micro-channel bonding and fabrication techniques for heat exchangers
- (3) Low dew-point dehumidifier development technology
- (4) Ultra-thin vapor chamber for next-generation semiconductor cooling
- (5) Compact inverter compressor-based dress care system development

RESEARCH PROJECTS

- (1) **Development of an AI-based Automated Design Tool for Data Center Direct Liquid Cooling (DLC) Systems** (PI) [2026]
- (2) **Development of a Direct Liquid Cooling (DLC) System for Reducing Power Consumption in Data Centers** (Lead) [2026-2029]
- (3) **Jet-Enhanced Immersion Cooling for Data Centers** (PI) [2025]
- (4) **KIMMPROP - Thermophysical Property iOS/Android App** (PI) [2025]
- (5) **PCHE for Liquid Hydrogen Supply System** (Lead) [2021-2026]
- (6) **Compact PCHE Design for Liquid Hydrogen** (Lead) [2022-2026]
- (7) **Immersion Cooling Waste Heat Utilization** (Lead) [2024-2028]
- (8) **300°C High-Temperature Heat Pump System** (Participant) [2023-2028]
- (9) **Chemisorption Heat Pump with Electrochemical Compressor** (Participant) [2021-2025]
- (10) **Next-Gen Alternative Refrigerant & HVAC Technology** (Participant) [2021-2025]
- (11) **Low-Charge Industrial Heat Pump** (Participant) [2022-2023]
- (12) **Water-Source Heating/Cooling Hybrid System** (Participant) [2022-2023]
- (13) **Heat Pipe HX Performance Enhancement** (Participant) [2022]
- (14) **Post H-class Gas Turbine Technology** (Participant) [2022]
- (15) **Compact Inverter Compressor Dress Care System** (Participant) [2022]
- (16) **Ultra-Thin Vapor Chamber Performance Enhancement** (Participant) [2022-2023]
- (17) **PCHE Freezing Test for Vehicle Heat Exchangers** (PI) [2025]
- (18) **BTMS Integrated A/Con System Evaluation** (Participant) [2025]
- (19) **3D Multiporous Cooling System for Ultra-high Heat Flux Applications** (Graduate Researcher) [2021]
- (20) **Smart protective suit with active cooling/ventilation and antivirus suit changing system** (Graduate Researcher) [2020]
- (21) **Next-Generation CPI Reliability Structural & Thermal Design** (Graduate Researcher) [2020]
- (22) **OHP High-Pressure Working-Fluid Charging Technology** (Graduate Researcher) [2020]
- (23) **Thermal optimization of MMU rear housing for reducing the weight** (Graduate Researcher) [2019-2020]
- (24) **Development of a Flexible Thin-Film Ultra-Thermal-Conductor** (Graduate Researcher) [2015-2020]
- (25) **Study on a smart space thermal management system** (Graduate Researcher) [2016-2018]

PROFESSIONAL ACTIVITIES

- (1) **Council Member**, Korean Society of Mechanical Engineers (KSME) [2026.04-Present]
- (2) **Committee Member - Thermal Management / Cooling**, SAREK Data Center Technology Committee [2026.03-Present]
- (3) **Regular Member**, Society of Air-Conditioning and Refrigerating Engineers of Korea (SAREK) [2022.04-Present]
- (4) **Regular Member**, Korean Society of Mechanical Engineers (KSME) [2021.05-Present]